

Ex - Find $\int \sin^3(x) \cos^2(x) dx$

$$= \int \sin^2(x) \cos^2(x) \sin(x) dx$$

$$= \int (1 - \cos^2(x)) \cdot \cos^2(x) \cdot \sin(x) dx$$

$$\cos^2(x) + \sin^2(x) = 1$$

$$u = \cos(x) \quad du = -\sin(x) dx \quad \frac{d}{dx} \cos(x) = -\sin(x)$$

$$= \int (1 - u^2) u^2 du \quad \cos^2(x) = u^2$$

See why this is

$$= \int (u^2 - u^4) du$$

$$= \frac{u^3}{3} - \frac{u^5}{5} + C = \frac{\cos^3(x)}{3} - \frac{\cos^5(x)}{5} + C$$

Case 1

$\sin^m$
m: odd

1.1

see pg 2

practice tonight

Candidate quiz problem - not necessarily type of problem quiz would be on

$$\int (x-1)e^{5x} dx$$

$$2 \int_1^t x e^{x^2} dx$$

$$\int x \sin x dx$$

Trigonometric Integrals:**Strategy 1:** Suppose you want to solve

$$\int \sin^m(x) \cos^n(x) dx.$$

- 1.1 1. If n is odd, save one cosine factor and use $\cos^2(x) = 1 - \sin^2(x)$ to express the remaining factors in terms of sine. $v = \sin x$
- 1.2 2. If m is odd, save one sine factor and use $\sin^2(x) = 1 - \cos^2(x)$ to express the remaining factors in terms of cosine. $v = \cos x$
- 1.3 3. If both m and n are even, use the half-angle identities

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x)), \quad \cos^2(x) = \frac{1}{2}(1 + \cos(2x)).$$

Example: Find the antiderivative

$$\int \sin^3(x) \cos(x) dx$$

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$$\int \sin^2(x) \cos^2(x) dx$$

$$= \int \underbrace{\frac{1}{2}(1 - \cos(2x))}_{\substack{\sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \\ \text{half angle}}} \cdot \underbrace{\frac{1}{2}(1 + \cos(2x))}_{\substack{\cos^2(x) = \frac{1}{2}(1 + \cos(2x)) \\ \text{half angle}}} dx$$

$$\stackrel{\text{distribute}}{=} \frac{1}{4} \int 1 - \cos^2(2x) dx = \frac{1}{4} \int \sin^2(2x) dx$$

$$\stackrel{\substack{\text{half angle, } \sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \\ 2x, n=2x}}{=} \frac{1}{4} \int \frac{1}{2} (1 - \cos(4x)) dx$$

$$= \frac{1}{8} \left[\int 1 dx - \int \cos(4x) dx \right]$$

$$\stackrel{\substack{\text{where is the } 4dx \text{ coming from?}}}{=} \frac{1}{8} \left[x - \frac{1}{4} \sin(4x) \right] + C$$

$$= \frac{1}{8} x - \frac{1}{32} \sin(4x) + C$$

Exercise: Use the identities

$$\sin(a) \sin(b) = \frac{1}{2} (\cos(a-b) - \cos(a+b))$$

to solve for the integral

$$\frac{2}{\pi} \int_0^{\pi} \sin(mx) \sin(nx) dx$$

where m, n are positive integers and:

1. $m \neq n$. ^{set arbitrary #s} $m=2$ $n=3$

$$\begin{aligned} \frac{2}{\pi} \int_0^{\pi} \sin(2x) \sin(3x) dx &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} (\cos(2x-3x) - \cos(2x+3x)) dx \\ &= \frac{2}{\pi} \cdot \frac{1}{2} \int_0^{\pi} \cos(-x) - \cos(5x) dx \\ &= \frac{2}{\pi} \cdot \frac{1}{2} \left[\sin(-x) \right]_0^{\pi} - \left[\sin(5x) \right]_0^{\pi} \\ &= \frac{2}{\pi} \cdot \frac{1}{2} [\sin(-\pi) - \sin(5\pi)] - [\sin(\pi) - \sin(5\pi)] = -0 - 0 = 0 \\ &\rightarrow \sin(-0) - \sin(5(0)) + \sin(0) - \sin(0) + 0 - 0 = 0 \end{aligned}$$

No matter the #, if $m \neq n$ it will always be 0

because $\sin(\pi) = 0$ and $\sin(0) = 0$ and this is true no matter what its multiplied by bc $0 \times = 0$

2. $m = n$.

when $m=n$, it will always output 1

why?

Quiz Wed. will be up to here